Quiz 4 Formula Sheet

1 Moment Generating Functions

1.1 Definition and Domain

The moment generating function (MGF) of X is

$$M_X(t) = E[e^{tX}] = \begin{cases} \displaystyle\int_{-\infty}^{\infty} e^{tx} f_X(x)\,dx, & X \text{ continuous}, \\ \displaystyle\sum_x e^{tx} P(X = x), & X \text{ discrete}. \end{cases}$$

Its domain consists of the values of t for which the integral or sum is finite. Always $M_X(0) = 1$. The moment and uniqueness properties below require finiteness on an open interval containing 0.

1.2 Recovering Moments

For $k = 1, 2, \ldots$,

$$M_X^{(k)}(0) = E[X^k].$$

In particular,

$$E[X] = M_X'(0), \qquad \mathrm{Var}(X) = M_X''(0) - \big(M_X'(0)\big)^2.$$

1.3 Scaling, Independent Sums, and Uniqueness

For a constant a, wherever $M_X(at)$ is finite,

$$M_{aX}(t) = M_X(at).$$

For independent $X_1, \ldots, X_n$, wherever their MGFs are finite,

$$M_{X_1+\cdots+X_n}(t) = \prod_{i=1}^{n} M_{X_i}(t).$$

If $M_X(t) = M_Y(t)$ on an open interval containing 0, then X and Y have the same distribution.

2 Standard Distributions

2.1 Exponential Density and CDF

For $X \sim \mathrm{Exp}(\lambda)$ with rate $\lambda > 0$, the density and CDF are

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0, \end{cases} \qquad F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-\lambda x}, & x \geq 0. \end{cases}$$

3 Standard MGFs

These MGFs use the parameterizations from Lectures 18–19.

3.1 Bernoulli Distribution

For $X \sim \text{Ber}(p)$, with $P(X = 1) = p$ and $P(X = 0) = 1 - p$,

$$M_X(t) = 1 - p + pe^t, \qquad t \in \mathbb{R}.$$

3.2 Poisson Distribution

For $X \sim \text{Pois}(\mu)$, where $\mu > 0$,

$$M_X(t) = \exp\big(\mu(e^t - 1)\big), \qquad t \in \mathbb{R}.$$

3.3 Gamma Distribution

For $X \sim \text{Gamma}(r, \lambda)$, where $r > 0$ is the shape and $\lambda > 0$ is the rate,

$$M_X(t) = \left(\frac{\lambda}{\lambda - t}\right)^r, \qquad t < \lambda.$$

Differentiating the MGF gives

$$E[X] = \frac{r}{\lambda}, \qquad E[X^2] = \frac{r(r+1)}{\lambda^2}.$$

$$\text{Var}(X) = \frac{r}{\lambda^2}.$$

3.4 Exponential Distribution

The exponential is the shape-1 gamma distribution. For $X \sim \text{Exp}(\lambda)$ with rate $\lambda > 0$,

$$M_X(t) = \frac{\lambda}{\lambda - t}, \qquad t < \lambda.$$

3.5 Normal Distribution

For $X \sim N(\mu, \sigma^2)$, where σ^2 denotes the variance,

$$M_X(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right), \qquad t \in \mathbb{R}.$$

For the standard normal $Z \sim N(0, 1)$,

$$M_Z(t) = e^{t^2/2}, \qquad t \in \mathbb{R}.$$

Notes

1. Suppose $T_i \overset{ind}{\sim} Exp(\lambda_i)$ for $i = 1, \ldots, n$. Let $X = \min(T_1, \ldots, T_n)$. Identify the distribution of X.

$$P(X > x) = \prod_{i=1}^{n} P(T_i > x) = \prod_{i=1}^{n} e^{-\lambda_i x} = e^{-(\lambda_1 + \cdots + \lambda_n)x}$$

$$X \sim Exp(\lambda_1 + \cdots + \lambda_n)$$

2. Suppose $T_i \overset{ind}{\sim} Exp(\lambda_i)$ for $i = 1, \ldots, n$. Let $Y = \min(T_1, \ldots, T_3)$. Find $f_Y(y)$, $E(Y)$ and $Var(Y)$. Note this is not a standard distribution.

Let $\Lambda = \lambda_1 + \lambda_2 + \lambda_3$

$Y \sim Exp(\Lambda)$

$f_Y(y) = \Lambda e^{-\Lambda y} \quad (y \geq 0)$

$E(Y) = \frac{1}{\Lambda}$

$Var(Y) = \frac{1}{\Lambda^2}$

$F_Y(y) = \prod_{i=1}^{3} (1 - e^{-\lambda_i y})$

$f_Y(y) = F_Y'(y)$

$E(Y) = \sum_i \frac{1}{\lambda_i} - \sum_{i<j} \frac{1}{\lambda_i + \lambda_j} + \frac{1}{\Lambda}$

$E(Y^2) = \sum_i \frac{2}{\lambda_i^2} - \sum_{i<j} \frac{2}{(\lambda_i + \lambda_j)^2} + \frac{2}{\Lambda^2}$

$Var(Y) = E(Y^2) - [E(Y)]^2$

3. Suppose $T \sim Exp(\lambda)$. Let $X = \lfloor T \rfloor$. Find:

a) $P(X = x)$

b) $E(X)$

c) $Var(X)$

a) $P(X = x) = P(x \le T < x+1)$

$= F(x+1) - F(x)$

$= e^{-\lambda x}(1 - e^{-\lambda})$ for $x = 0, 1, 2, \ldots$

So it's a geom distribution (counting failures) with $p = 1 - e^{-\lambda}$

b) $E(X) = \frac{1-p}{p} = \frac{e^{-\lambda}}{1-e^{-\lambda}} = \frac{1}{e^{\lambda} - 1}$

c) $Var(X) = \frac{1-p}{p^2} = \frac{e^{\lambda}}{(e^{\lambda}-1)^2}$

4. Find $f_Y(y)$ and identify its distribution for each chance of variable.

a) Suppose $X \sim Unif(0,1)$. Let $Y = -\log X$.

$$P(-\log X \le y) = P(X \ge e^{-y})$$

$$= 1 - e^{-y}$$

$Y \sim Exp(1)$

b) Suppose $X \sim Exp(\lambda)$. Let $Y = cX$ for $c > 0$.
(You can prove this result in 3 different ways, PDF, CDF, and MGF.)

$Y \sim Exp\left(\frac{\lambda}{c}\right)$

① CDF:

$$P(cX \le y) = 1 - e^{-\left(\frac{\lambda}{c}\right)y}$$

② PDF:

$$f_Y(y) = \frac{1}{c} f_X\left(\frac{y}{c}\right) = \frac{\lambda}{c} e^{-\left(\frac{\lambda}{c}\right)y}$$

③ MGF:

$$M_Y(t) = M_X(ct) = \frac{\lambda}{\lambda - ct} = \frac{\lambda/c}{\lambda/c - t}$$

c) Suppose $X \sim Unif(0,1)$. Let $Y = -\log\sqrt{X}$.
d) Suppose $X \sim Gamma(r,\lambda)$. Let $Y = cX$ for $c > 0$.

c) $Y = -\log\sqrt{X} = \frac{1}{2}(-\log X)$

$P(Y \le y) = P(X \ge e^{-2y}) = 1 - e^{-2y}$

$Y \sim Exp(2)$

d) $M_Y(t) = \left(\frac{\lambda}{\lambda - ct}\right)^r = \left(\frac{\lambda/c}{\lambda/c - t}\right)^r$

$Y \sim Gamma\left(r, \frac{\lambda}{c}\right)$

e) Suppose $Z \sim N(0,1)$. Let $Y = a + bX$.
f) Suppose $Z \sim N(0,1)$. Find a transformation $X = g(Z)$ such that $X \sim N(\mu, \sigma^2)$

e) $M_Y(t) = E\left[e^{t(a+bZ)}\right]$

$= e^{at} M_Z(bt)$

$= \exp\left(at + \frac{b^2t^2}{2}\right)$

$Y \sim N(a, b^2)$

f) $X = g(Z) = \mu + \sigma Z \quad (\sigma > 0)$

$M_X(t) = e^{\mu t} e^{\sigma^2 t^2/2}$

which is $N(\mu^2, \sigma^2)$ MGF

g) Suppose $X \sim N(0,1)$. Let $Y = X^2$.
h) Suppose $X \sim N(0,\sigma^2)$. Let $Y = X^2$.

g) $M_Y(t) = E[e^{tZ^2}] = \int_{-\infty}^{\infty} e^{tz^2} \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz$

$= (1-2t)^{-\frac{1}{2}} \qquad t < \frac{1}{2}$

$(1-2t)^{-\frac{1}{2}} = \left(\frac{1/2}{1/2-t}\right)^{\frac{1}{2}}$

$Y \sim \text{Gamma}\left(\frac{1}{2}, \frac{1}{2}\right)$

h) $X = \sigma Z$

$Y = \sigma^2 Z^2$

$Y \sim \text{Gamma}\left(\frac{1}{2}, \frac{1}{2\sigma^2}\right)$

5. Suppose $X \sim Exp(\lambda)$. Let $Y = cX, c > 0$.

a) Find $M_Y(t)$.

b) Identity the distribution of Y.

a) $M_Y(t) = M_X(ct) = \frac{\lambda}{\lambda - ct}$ $\quad t < \frac{\lambda}{c}$

b) $= \frac{\lambda/c}{\lambda/c - t}$

$Y \sim Exp(\frac{\lambda}{c})$

6. Suppose $M_Y(t) = \frac{1}{8}e^{3t} + \frac{3}{8}e^{4t} + \frac{3}{8}e^{5t} + \frac{1}{8}e^{6t}$. Find $E(Y)$ and $Var(Y)$. Hint $Y = X + c$.

$M_Y(t) = e^{3t}\left(\frac{1+e^t}{2}\right)^3$

So $Y = X+3$ with $X \sim Bin\left(3, \frac{1}{2}\right)$

$E(Y) = 3 \cdot \frac{1}{2} + 3 = 4.5$

$Var(Y) = 3 \cdot \frac{1}{4} = 0.75$

Stat 134 · Quiz 4 Review and Predicted Quiz

MGFs · change of variable · Lectures 17–19. Tags: [§x.y] = that item on the Quiz 4 sheet, [off-sheet] = know it yourself.

Part 1. The formula sheet, explained from scratch

You already know (Quizzes 1–3) what a pmf, a density, a CDF, an expectation and a variance are, what independence means, and the named distributions. Everything new for Quiz 4 is built from those. Read this part once, slowly; the questions in Part 2 only use what is here.

[§1.1] What a moment generating function is

From Quiz 2 you know how to average a function of a random variable: $\mathbb{E}[g(X)] = \sum_x g(x)P(X = x)$ for a discrete X, or $\int g(x)f_X(x)\,dx$ for a continuous one. The moment generating function (MGF) is nothing more than this average for one particular choice of function, $g(x) = e^{tx}$, where t is a fixed number that you get to choose:

$$M_X(t) = \mathbb{E}[e^{tX}] = \begin{cases} \sum_x e^{tx}P(X = x) & X \text{ discrete,} \\ \int e^{tx}f_X(x)\,dx & X \text{ continuous.} \end{cases}$$

For each value of t you get one number, so M_X is a function of t; the randomness has been averaged away. Example: a fair coin, $X = 1$ for heads and 0 for tails, has $M_X(t) = \frac{1}{2}e^0 + \frac{1}{2}e^t = \frac{1}{2} + \frac{1}{2}e^t$. Notice that for a discrete variable the MGF is literally the pmf written out: each possible value x contributes a term $P(X = x)\,e^{tx}$, so the coefficient in front of e^{tx} *is* $P(X = x)$. That is how you read a pmf off an MGF.

The *domain* is the set of t for which the average is finite. For a variable with finitely many values it is all of $\mathbb{R}$. For an exponential variable the integrand is $e^{tx} \cdot \lambda e^{-\lambda x} = \lambda e^{-(\lambda - t)x}$, which only decays (and so only has a finite integral) when $t < \lambda$; for $t \geq \lambda$ the average is infinite. Some heavy-tailed variables (for instance one with density $1/y^2$ on $y > 1$) have an infinite average for every $t > 0$ and so do not have an MGF at all. Finally, at $t = 0$ we are averaging $e^0 = 1$, so $M_X(0) = 1$ for every random variable. This is the fastest way to reject a wrong multiple-choice option.

[§1.2] Why it is called "moment generating": derivatives at 0 are moments

The kth *moment* of X is $\mathbb{E}[X^k]$; the first moment is the mean and the second gives the variance through $\mathrm{Var}X = \mathbb{E}X^2 - (\mathbb{E}X)^2$ (Quiz 2 sheet). The MGF stores all of them. Expand the exponential as a series, $e^{tX} = 1 + tX + \frac{t^2X^2}{2!} + \frac{t^3X^3}{3!} + \cdots$, and take expectations term by term:

$$M_X(t) = 1 + t\,\mathbb{E}[X] + \frac{t^2}{2!}\mathbb{E}[X^2] + \frac{t^3}{3!}\mathbb{E}[X^3] + \cdots.$$

Differentiating once and setting $t = 0$ kills every term except $\mathbb{E}[X]$; differentiating twice leaves $\mathbb{E}[X^2]$; and so on. That is the sheet's statement $M_X^{(k)}(0) = \mathbb{E}[X^k]$, and in particular

$$\mathbb{E}[X] = M_X'(0), \qquad \mathrm{Var}(X) = M_X''(0) - \left(M_X'(0)\right)^2.$$

Check with the coin: $M'_X(t) = \frac{1}{2}e^t$, so $M'_X(0) = \frac{1}{2} = \mathbb{E}X$. On a question you can either differentiate the given formula or, often faster, expand it in powers of t and read off: $\mathbb{E}[X^k]$ equals $k!$ times the coefficient of t^k.

[§1.3] Three rules: scaling, independent sums, uniqueness

Scaling. If you multiply X by a constant a, then $e^{t(aX)} = e^{(at)X}$, so averaging gives $M_{aX}(t) = M_X(at)$: the new MGF is the old one with t replaced by at. The sheet stops here, but the same one-line argument handles adding a constant: $e^{t(X+b)} = e^{tb}\,e^{tX}$ and e^{tb} is not random, so it comes out of the average, $M_{X+b}(t) = e^{bt}M_X(t)$. Together, $M_{aX+b}(t) = e^{bt}M_X(at)$ **[off-sheet]**. Adding a constant multiplies the MGF by e^{bt}; it never adds b to it.

Independent sums. For two variables, $e^{t(X+Y)} = e^{tX}e^{tY}$. From Quiz 3 you know that the expectation of a product equals the product of expectations *when the factors are independent*: $\mathbb{E}[e^{tX}e^{tY}] = \mathbb{E}[e^{tX}]\mathbb{E}[e^{tY}]$. Hence $M_{X+Y}(t) = M_X(t)M_Y(t)$ for independent X, Y, and for n independent variables the MGF of the sum is the product of the n MGFs (all evaluated at the same t). If independence is not stated, this step is simply not available, which is what "not enough information" means on the quiz. Two things that look similar but are different: $M_X(t)^2$ is the MGF of $X + X'$ where X' is an independent copy of X, while $M_{2X}(t) = M_X(2t)$ is the MGF of $2X$.

Uniqueness. Different distributions always have different MGFs. So if you compute the MGF of some variable and it turns out to be, say, $(\frac{3}{3-t})^2$, you may conclude that the variable *is* Gamma$(2, 3)$, because that is the only distribution with that MGF. This is how every "name the distribution" question works: compute, match the shape to §3, read off the parameters. Note the contrast: two distributions can share a mean and a variance (Poisson(1) and Exp(1) both have mean 1 and variance 1) without being the same, but they cannot share an MGF.

[§2.1] The exponential distribution

$X \sim$Exp(λ) is the continuous waiting time for the first arrival in a Poisson process with rate λ (midterm material). Its density is $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$, its CDF is $F_X(x) = P(X \leq x) = 1 - e^{-\lambda x}$, and so its tail is $P(X > x) = e^{-\lambda x}$. Its mean is $1/\lambda$ and its variance $1/\lambda^2$. It is on this sheet because it is the input to most change-of-variable questions and because the exponential MGF (§3.4) is computed from this density.

[§3.1]–[§3.5] Where the five MGFs come from, and how to recognise them

Bernoulli(p): $M(t) = 1-p+pe^t$. Two values: 0 with probability $1-p$ contributes $(1-p)e^0$, and 1 with probability p contributes pe^t. Multiply n independent copies (rule 1.3) and you get the Binomial(n, p) MGF $(1-p+pe^t)^n$ **[off-sheet]**, because a binomial count is a sum of n independent Bernoulli indicators. Recognise: a two-term expression $q + pe^t$ with $q + p = 1$, possibly raised to a power n.

Poisson(μ): $M(t) = \exp\left(\mu(e^t - 1)\right)$. From the definition, $\sum_{k\geq 0} e^{tk}\, e^{-\mu}\frac{\mu^k}{k!} = e^{-\mu}\sum_k \frac{(\mu e^t)^k}{k!} = e^{-\mu}e^{\mu e^t}$, using the series $\sum u^k/k! = e^u$ with $u = \mu e^t$. Recognise: e raised to "a constant times $(e^t - 1)$"; the constant is μ. Two independent Poissons add: the exponents add, so Pois(μ_1)+Pois$(\mu_2) \sim$Pois$(\mu_1 + \mu_2)$.

Exponential(λ)**:** $M(t) = \frac{\lambda}{\lambda - t}$ **for** $t < \lambda$**.** From the definition with the density in §2.1: $\int_0^\infty e^{tx}\lambda e^{-\lambda x}dx = \lambda\int_0^\infty e^{-(\lambda-t)x}dx = \frac{\lambda}{\lambda-t}$, valid only when $\lambda - t > 0$. Recognise: a single fraction "rate over rate minus t". Useful consequence: $M_{cX}(t) = \frac{\lambda}{\lambda - ct} = \frac{\lambda/c}{\lambda/c - t}$, so multiplying an exponential by c gives an exponential with rate λ/c.

Gamma(r,λ)**:** $M(t) = \left(\frac{\lambda}{\lambda-t}\right)^r$ **for** $t < \lambda$**.** A Gamma(r,λ) variable is the waiting time for the rth arrival, i.e. the sum of r independent Exp(λ) waits; by rule 1.3 its MGF is the exponential MGF raised to the power r. The first parameter r is the *shape* (how many exponentials were added) and the second, λ, is the *rate*. The sheet also prints the moments you get by differentiating: $\mathbb{E}X = \frac{r}{\lambda}$, $\mathbb{E}X^2 = \frac{r(r+1)}{\lambda^2}$, $\mathrm{Var}X = \frac{r}{\lambda^2}$. Recognise: a fraction $\frac{\lambda}{\lambda-t}$ raised to a power; the same thing is often written $(1 - t/\lambda)^{-r}$, so $(1-3t)^{-2}$ means $\lambda = \frac{1}{3}$, $r = 2$. Independent gammas with the *same* rate add their shapes; with different rates the product of MGFs is not a gamma.

Normal(μ,σ^2)**:** $M(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$**; standard normal** $M_Z(t) = e^{t^2/2}$**.** For Z, complete the square in the exponent of $\int e^{tz}\frac{1}{\sqrt{2\pi}}e^{-z^2/2}dz$: $tz - \frac{z^2}{2} = -\frac{(z-t)^2}{2} + \frac{t^2}{2}$, and the remaining integral is a normal density that integrates to 1, leaving $e^{t^2/2}$. For a general normal write $X = \mu + \sigma Z$ and apply shift-and-scale: $M_X(t) = e^{\mu t}M_Z(\sigma t) = e^{\mu t}e^{\sigma^2 t^2/2}$. Recognise: e raised to a polynomial in t of degree two. The coefficient of t is the mean; the coefficient of t^2 is *half* the variance (so e^{2t+8t^2} is $N(2,16)$). Independent normals add means and variances, because exponents add.

Change of variable (Pitman 4.4, not on the sheet)

The question: X has a known density and $Y = g(X)$ is a function of it. What is the distribution of Y?

The CDF method always works and uses only midterm skills. Compute $F_Y(y) = P(Y \le y) = P(g(X) \le y)$, solve the inequality $g(X) \le y$ for X (if g is decreasing the inequality flips), and evaluate with F_X. If a density is wanted, differentiate. Example: $U \sim$Unif$(0,1)$ and $Y = -\ln U$. Then $P(Y > y) = P(-\ln U > y) = P(\ln U < -y) = P(U < e^{-y}) = e^{-y}$, which is the Exp$(1)$ tail, so $Y \sim$Exp(1).

The density formula is a shortcut when g is one-to-one (strictly increasing or strictly decreasing) on the range of X. The probability that Y falls in a tiny interval of width dy equals the probability that X falls in the matching tiny interval of width dx: $f_Y(y)\,dy = f_X(x)\,dx$. Dividing,

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \left|\frac{dx}{dy}\right| \qquad \text{for } y \text{ in the range of } Y.$$

In words: the density of X at the point that maps to y, times the factor $|dx/dy|$ that measures how much g stretches or squeezes lengths there. The absolute value keeps the density positive when g is decreasing. Always write down the range of Y first; many wrong options are the right formula on the wrong interval. For the linear case $Y = aX + b$ this gives $f_Y(y) = \frac{1}{|a|}f_X\left(\frac{y-b}{a}\right)$.

Two-to-one functions such as $Y = X^2$ when X can be negative: two values $\pm\sqrt{y}$ map to the same y, so add the two contributions, $f_Y(y) = \frac{f_X(\sqrt{y}) + f_X(-\sqrt{y})}{2\sqrt{y}}$. The CDF method does this automatically: $P(X^2 \le y) = P(-\sqrt{y} \le X \le \sqrt{y})$.

Probabilities and means of Y **do not need** f_Y**.** For $P(Y > c)$ rewrite the event in terms of X and use F_X. For $\mathbb{E}[Y]$ use $\mathbb{E}[g(X)] = \int g(x)f_X(x)\,dx$, or known moments of X (for instance $\mathbb{E}[X^2] = \mathrm{Var}X + (\mathbb{E}X)^2$, or $M_X''(0)$).

Discrete outputs. If $g(X)$ takes integer values (for example $\lfloor X \rfloor$), there is no density and no Jacobian: compute $P(Y = k)$ as the probability that X lands in the set of x with $g(x) = k$, using F_X.

Named facts you should know by heart (all off-sheet)

$-\frac{1}{\lambda}\ln U \sim \text{Exp}(\lambda)$ for $U \sim \text{Unif}(0,1)$. $c\cdot\text{Exp}(\lambda) \sim \text{Exp}(\lambda/c)$ and $c\cdot\text{Gamma}(r,\lambda) \sim \text{Gamma}(r,\lambda/c)$. $\mu + \sigma Z \sim N(\mu, \sigma^2)$. The minimum of independent $\text{Exp}(\lambda_i)$'s is $\text{Exp}(\sum \lambda_i)$ because $P(\min > x) = \prod e^{-\lambda_i x}$. $\lfloor \text{Exp}(\lambda) \rfloor$ is geometric on $\{0, 1, \dots\}$ with $p = 1 - e^{-\lambda}$. Z^2 has MGF $(1-2t)^{-1/2}$, i.e. $\text{Gamma}(\frac{1}{2}, \frac{1}{2})$. Means and variances of Bin, Geom and Unif from the Quiz 2–3 sheets.

Part 2. Predicted quiz

Question 1 (change of variable)

Let $X \sim \text{Exp}(2)$ and $U \sim \text{Unif}(0,1)$, independent.

(a) $Y = 3X$. Select $M_Y(t)$: (i) $\frac{2}{2-t}$ (ii) $\frac{2}{2-3t}$, $t < \frac{2}{3}$ (iii) $\left(\frac{2}{2-t}\right)^3$ (iv) $\frac{6}{6-t}$ (v) None. Name the distribution of Y.

$M_{3X}(t) = M_X(3t)$ [§1.3] $= \frac{2}{2-3t}$ [§3.4], needs $3t < 2$: **(ii)**. Divide top and bottom by 3: $\frac{2/3}{2/3-t}$, the Exp MGF with rate $\frac{2}{3}$, so $Y \sim$ **Exp**$(2/3)$ [§1.3] uniqueness. (Scaling by c divides the rate by c; mean becomes $3 \cdot \frac{1}{2}$.) (iii) is $X_1 + X_2 + X_3$; (iv) is $X/3$.

(b) $W = -\frac{1}{2}\ln U$. Name the distribution of W and compute $P(W > 1)$ to three decimals.

Tail: $P(W > w) = P(\ln U < -2w) = P(U < e^{-2w}) = e^{-2w}$, the Exp tail with $\lambda = 2$ [§2.1]. So $W \sim$ **Exp**(2) and $P(W > 1) = e^{-2} \approx$ **0.135**. Trap: the $\frac{1}{2}$ in front is $1/\lambda$, not λ.

(c) $V = X^2$. Select the density of V for $v > 0$: (i) $2e^{-2v^2}$ (ii) $\frac{e^{-2\sqrt{v}}}{\sqrt{v}}$ (iii) $\frac{2e^{-2\sqrt{v}}}{\sqrt{v}}$ (iv) $4\sqrt{v}\,e^{-2\sqrt{v}}$ (v) None. Then compute $P(V > 4)$ and $\mathbb{E}[V]$.

One-to-one on $x > 0$ [off-sheet]: $x = \sqrt{v}$, $|dx/dv| = \frac{1}{2\sqrt{v}}$, so $f_V(v) = f_X(\sqrt{v})\cdot\frac{1}{2\sqrt{v}} = 2e^{-2\sqrt{v}}\cdot\frac{1}{2\sqrt{v}} = \frac{e^{-2\sqrt{v}}}{\sqrt{v}}$: **(ii)**. (iii) forgot the Jacobian, (i) plugged v in for x, (iv) put $g'(x) = 2x$ on top instead of the bottom.

$P(V > 4) = P(X > 2) = e^{-4} \approx$ **0.018** [§2.1]. $\mathbb{E}V = \mathbb{E}X^2 = M_X''(0)$ [§1.2]: $M_X = \frac{2}{2-t}$, $M_X'' = \frac{4}{(2-t)^3}$, so $\mathbb{E}V = \frac{4}{8} =$ **0.5** $(= \text{Var}X + (\mathbb{E}X)^2 = \frac{1}{4} + \frac{1}{4})$.

(d) $T = \lfloor X \rfloor$ (round down). Select $P(T = k)$, $k = 0, 1, \dots$: (i) e^{-2k} (ii) $e^{-2k}(1 - e^{-2})$ (iii) $2e^{-2k}$ (iv) $(1 - e^{-2})^k e^{-2}$ (v) None. Name the distribution and give $\mathbb{E}[T]$.

$\{T = k\} = \{k \le X < k + 1\}$, so $P(T = k) = F(k+1) - F(k) = e^{-2k} - e^{-2(k+1)} = e^{-2k}(1 - e^{-2})$ [§2.1]: **(ii)**. With $q = e^{-2}$, $p = 1 - q$ this is $q^k p$: **Geometric on** $\{0, 1, \dots\}$ with $p = 1 - e^{-2}$ [off-sheet]. $\mathbb{E}T = \frac{q}{p} = \frac{e^{-2}}{1-e^{-2}} \approx$ **0.157**. (Round *up* instead: Geom on $\{1, 2, \dots\}$, mean $1/p$.)

Question 2 (reading an MGF)

(a) $M_Y(t) = \frac{1}{8}e^{3t} + \frac{3}{8}e^{4t} + \frac{3}{8}e^{5t} + \frac{1}{8}e^{6t}$. Find $P(Y = 5)$, $\mathbb{E}[Y]$, $\mathrm{Var}(Y)$.

Discrete MGF = pmf list [§1.1]: $P(Y = 5) = \mathbf{3/8}$. $\mathbb{E}Y = \frac{3+12+15+6}{8} = \mathbf{4.5}$; $\mathbb{E}Y^2 = \frac{9+48+75+36}{8} = 21$; $\mathrm{Var}Y = 21 - 4.5^2 = \mathbf{0.75}$. (Same numbers as $M'(0)$, $M''(0) - M'(0)^2$ [§1.2].)

(b) Write $Y = X + c$ with X a named distribution. Select: (i) $X \sim \mathrm{Bin}(3, \frac{1}{2})$, $c = 3$ (ii) $X \sim \mathrm{Bin}(6, \frac{1}{2})$, $c = 0$ (iii) $X \sim \mathrm{Pois}(3)$, $c = 3$ (iv) None.

Factor the smallest exponential: $M_Y(t) = e^{3t}(\frac{1}{8} + \frac{3}{8}e^t + \frac{3}{8}e^{2t} + \frac{1}{8}e^{3t}) = e^{3t}(\frac{1}{2} + \frac{1}{2}e^t)^3$. $e^{3t}M_X(t)$ is the MGF of $X + 3$ [off-sheet], and $(\frac{1}{2} + \frac{1}{2}e^t)^3$ is three Bernoulli($\frac{1}{2}$)'s multiplied [§3.1][§1.3] = $\mathrm{Bin}(3, \frac{1}{2})$: **(i)**. Check: $\mathbb{E}Y = 1.5 + 3$, $\mathrm{Var}Y = 3 \cdot \frac{1}{4} = 0.75$.

(c) $M_X(t) = e^{2t+8t^2}$. Name the distribution; give the SD and $\mathbb{E}[X^2]$.

Match $e^{\mu t + \sigma^2 t^2/2}$ [§3.5]: $\mu = 2$, $\sigma^2/2 = 8 \Rightarrow \sigma^2 = 16$. $N(2, 16)$, SD **4** (not $\sqrt{8}$). $\mathbb{E}X^2 = \mathrm{Var} + \mu^2 = \mathbf{20}$.

(d) $M_X(t) = (1 - 3t)^{-2}$, $t < \frac{1}{3}$. Name the distribution; give $\mathbb{E}X$, $\mathrm{Var}X$, $\mathbb{E}X^2$.

$(1 - t/\lambda)^{-r} = (\frac{\lambda}{\lambda - t})^r$ with $\lambda = \frac{1}{3}$, $r = 2$: **Gamma**$(2, \frac{1}{3})$ [§3.3]. $\mathbb{E}X = \frac{r}{\lambda} = \mathbf{6}$, $\mathrm{Var} = \frac{r}{\lambda^2} = \mathbf{18}$, $\mathbb{E}X^2 = \frac{r(r+1)}{\lambda^2} = \mathbf{54}$ [§3.3]. Shape first, rate second.

(e) Select all that are always true: (i) $M_X(0) = 1$ (ii) $M_{2X}(t) = M_X(t)^2$ (iii) $M_{X+3}(t) = M_X(t) + 3$ (iv) $\mathrm{Var}X = M_X''(0)$ (v) same mean and variance ⇒ same distribution (vi) same MGF near 0 ⇒ same distribution (vii) $X \sim \mathrm{Exp}(\lambda)$: $M_X(t)$ finite for all t.

(i), (vi) only. (ii) $M_{2X}(t) = M_X(2t)$ [§1.3]; $M_X(t)^2$ is $X + X'$ for an independent copy. (iii) $e^{3t}M_X(t)$ [off-sheet]. (iv) $M''(0) = \mathbb{E}X^2$; subtract $M'(0)^2$ [§1.2]. (v) Pois(1) vs Exp(1). (vii) only $t < \lambda$ [§3.4].

Question 3 (building an MGF)

X_1, X_2, X_3 are independent $\mathrm{Exp}(4)$; $Y \sim N(1, 9)$ is independent of them.

(a) $S = X_1 + X_2 + X_3$. Select $M_S(t)$: (i) $\frac{4}{4-t}$ (ii) $\frac{12}{12-t}$ (iii) $(\frac{4}{4-t})^3$, $t < 4$ (iv) $(\frac{4}{4-3t})^3$ (v) None. Name S and compute $\mathbb{E}[S^2]$.

Independent: multiply [§1.3] three copies of $\frac{4}{4-t}$ [§3.4]: **(iii)**. That is [§3.3] with $r = 3, \lambda = 4$: $S \sim$ **Gamma**$(3, 4)$. $\mathbb{E}S^2 = \frac{r(r+1)}{\lambda^2} = \frac{12}{16} = \mathbf{0.75}$ [§3.3]. Trap: (ii) Exp(12) is the *minimum* (tails multiply: $e^{-4x}e^{-4x}e^{-4x}$), not the sum.

(b) $T = 3X_1 - 2$. Select $M_T(t)$: (i) $\frac{4}{4-3t} - 2$ (ii) $e^{-2t}\frac{4}{4-3t}$, $t < \frac{4}{3}$ (iii) $e^{2t}\frac{4}{4-3t}$ (iv) $e^{-2t}\frac{12}{12-t}$ (v) None. Give $\mathbb{E}T$ and $\mathrm{Var}T$.

$M_{aX+b}(t) = e^{bt}M_X(at)$ [off-sheet] with $a = 3, b = -2$: $e^{-2t}\frac{4}{4-3t}$ [§3.4], domain $3t < 4$: **(ii)**. Check

$t = 0$ gives 1. $\mathbb{E}T = 3 \cdot \frac{1}{4} - 2 = \mathbf{-1.25}$, $\mathrm{Var}T = 9 \cdot \frac{1}{16} = \mathbf{0.5625}$.

(c) $R = 2Y + X_1$. Select $M_R(t)$ for $t < 4$: (i) $e^{2t+18t^2}\frac{4}{4-t}$ (ii) $e^{t+9t^2/2} + \frac{4}{4-t}$ (iii) $e^{2t+9t^2}\frac{4}{4-t}$ (iv) $e^{2t+36t^2}\frac{4}{4-t}$ (v) None. Give $\mathbb{E}R$, $\mathrm{Var}R$. Is R normal?

$M_{2Y}(t) = M_Y(2t) = e^{2t+\frac{9}{2}(2t)^2} = e^{2t+18t^2}$ [§1.3][§3.5] (so $2Y \sim N(2, 36)$). Independent, so multiply by $\frac{4}{4-t}$ [§3.4]: **(i)**. (iii) forgot to square the 2. $\mathbb{E}R = 2 + \frac{1}{4} = \mathbf{2.25}$, $\mathrm{Var}R = 36 + \frac{1}{16} = \mathbf{36.0625}$. Not normal: the MGF is not $e^{\text{quadratic}}$, so by uniqueness [§1.3] it is not $N(\cdot, \cdot)$.

(d) $P \sim$Pois(2) and $Q \sim$Pois(3), independent. Name $P + Q$ and compute $P(P + Q = 0)$. Then: A and B have $\mathbb{E}A = 2$, $\mathbb{E}B = 3$, nothing said about independence. $M_{A+B}(t)$? (enter -1 if not enough information)

$e^{2(e^t-1)}e^{3(e^t-1)} = e^{5(e^t-1)}$ [§3.2][§1.3]: **Pois**(5), $P(=0) = e^{-5} \approx \mathbf{0.0067}$. M_{A+B}: $\mathbf{-1}$; the product rule [§1.3] needs independence (and $\mathbb{E}[A + B] = 5$ is still fine by linearity).

(e) $I_1, \ldots, I_{10}$ independent Bernoulli(0.3), $X = \sum I_k$. Compute $\mathbb{E}[2^X]$ to two decimals.

$M_X(t) = (0.7 + 0.3e^t)^{10}$ [§3.1][§1.3]. $\mathbb{E}[2^X] = \mathbb{E}[e^{(\ln 2)X}] = M_X(\ln 2)$ [§1.1] $= (0.7 + 0.6)^{10} = 1.3^{10} \approx \mathbf{13.79}$. (Not $2^{\mathbb{E}X} = 8$.)

Part 3. Variants to expect (one line each)

Q1-type

- $U \sim$Unif(0, 1): $-\ln U \sim$Exp(1); $-\frac{1}{\lambda}\ln U \sim$Exp($\lambda$); $U^{1/n}$ has density ny^{n-1} on $(0, 1)$; $1/U$ has density $1/y^2$ on $y > 1$ and infinite mean (so no MGF [§1.1]).
- $X \sim$Gamma(r, λ): $cX \sim$Gamma($r, \lambda/c$) (shape unchanged) [§3.3][§1.3].
- $Z \sim N(0, 1)$: $a + bZ \sim N(a, b^2)$; Z^2 has density $\frac{e^{-y/2}}{\sqrt{2\pi y}}$ and MGF $(1-2t)^{-1/2} =$ Gamma($\frac{1}{2}, \frac{1}{2}$) [§3.3]; $\mathbb{E}Z^4 = 3$; $|Z|$ has density $2\varphi(y)$.
- $X \sim N(0, \sigma^2)$: $X^2 \sim$Gamma($\frac{1}{2}, \frac{1}{2\sigma^2}$), mean σ^2.
- $T_i \sim$Exp(λ_i) independent: min $\sim$Exp($\sum \lambda_i$), $P(T_1$ first$) = \lambda_1 / \sum \lambda_i$. Max of 3 iid Exp($\lambda$): $F = (1 - e^{-\lambda y})^3$, $\mathbb{E} = \frac{1}{3\lambda} + \frac{1}{2\lambda} + \frac{1}{\lambda}$, not a named law.
- $X \sim$Exp(λ), $Y = \sqrt{X}$: $2\lambda y e^{-\lambda y^2}$; $Y = e^X$: $\lambda y^{-\lambda-1}$, $y > 1$.

Q2-type

- $(0.25 + 0.75e^t)^8$: Bin(8, 0.75), mean 6, var 1.5, $P(X = 8) = 0.75^8$. $0.3e^{-2t} + 0.7e^{5t}$: two-point pmf, mean 2.9.
- e^{4e^t-4}: Pois(4), $P(X = 0) = e^{-4}$. e^{4e^t} alone: not an MGF ($M(0) \neq 1$).
- $(\frac{2}{2-t})^3$: Gamma(3, 2), mean 1.5, $\mathbb{E}X^2 = 3$. $\frac{5}{5-t}$: Exp(5); M_{3X} finite for $t < \frac{5}{3}$.
- e^{3t+2t^2}: $N(3, 4)$. $e^{t^2/2}$: $\mathbb{E}X^3 = 0$, $\mathbb{E}X^4 = 3$ (series: $e^{t^2/2} = 1 + \frac{t^2}{2} + \frac{t^4}{8} + \ldots$, $\mathbb{E}X^4 = 4! \cdot \frac{1}{8}$).

- $\frac{1}{1-t^2}$: no name; series $1 + t^2 + t^4$ gives mean 0, variance 2, $\mathbb{E}X^4 = 24$.
- $\frac{\lambda}{\lambda - t}$ by derivatives: $M' = \lambda(\lambda - t)^{-2} \to \frac{1}{\lambda}$; $M'' = 2\lambda(\lambda - t)^{-3} \to \frac{2}{\lambda^2}$; Var $= \frac{1}{\lambda^2}$.

Q3-type

- Closed under independent sums: Bin (same p), Pois, Gamma (same λ), Normal. Not closed: Exp(1) + Exp(2) (the product $\frac{1}{1-t} \cdot \frac{2}{2-t}$ is not a gamma), Unif + Unif, Bin with different p. Means and variances still add.
- $3N_1 - 2N_2 + 1$ with $N_1 \sim N(1, 4)$, $N_2 \sim N(2, 9)$: $N(0, 72)$; the -2 contributes $+36$ to the variance.
- $X + X'$ (independent copy) vs $2X$ for $X \sim N(0, 1)$: $e^{t^2} = N(0, 2)$ vs $e^{2t^2} = N(0, 4)$.
- $\mathbb{E}[e^X] = M_X(1)$: for $N(\mu, \sigma^2)$ it is $e^{\mu + \sigma^2/2}$; $\mathbb{E}[a^X] = M_X(\ln a)$.
- Centering: $M_{X-\mu}(t) = e^{-\mu t} M_X(t)$; its second derivative at 0 is Var$X$.
- From the definition [§1.1]: X uniform on $\{1, 2, 3\}$: $\frac{1}{3}(e^t + e^{2t} + e^{3t})$; Unif$(a, b)$: $\frac{e^{tb} - e^{ta}}{t(b-a)}$; Geom$\{1, 2, ..\}$: $\frac{pe^t}{1-(1-p)e^t}$, $t < -\ln(1-p)$.

Traps

- Any option with $M(0) \neq 1$ or $M'(0) \neq$ the known mean is wrong.
- $M_{2X}(t) = M_X(2t)$, not $M_X(t)^2$. $M_{X+b} = e^{bt} M_X$, not $M_X + b$.
- Products of MGFs and additivity of variance need independence; unstated ⇒ "not enough information."
- Normal: t^2 coefficient $= \sigma^2/2$. Gamma: shape first, rate second; domain $t < \lambda$ (after scaling by a: $at < \lambda$).
- Same mean and variance ≠ same distribution; same MGF = same distribution.
- Change of variable: Jacobian $|dx/dy|$, density never negative, state the range; sums multiply MGFs, minima multiply tails.